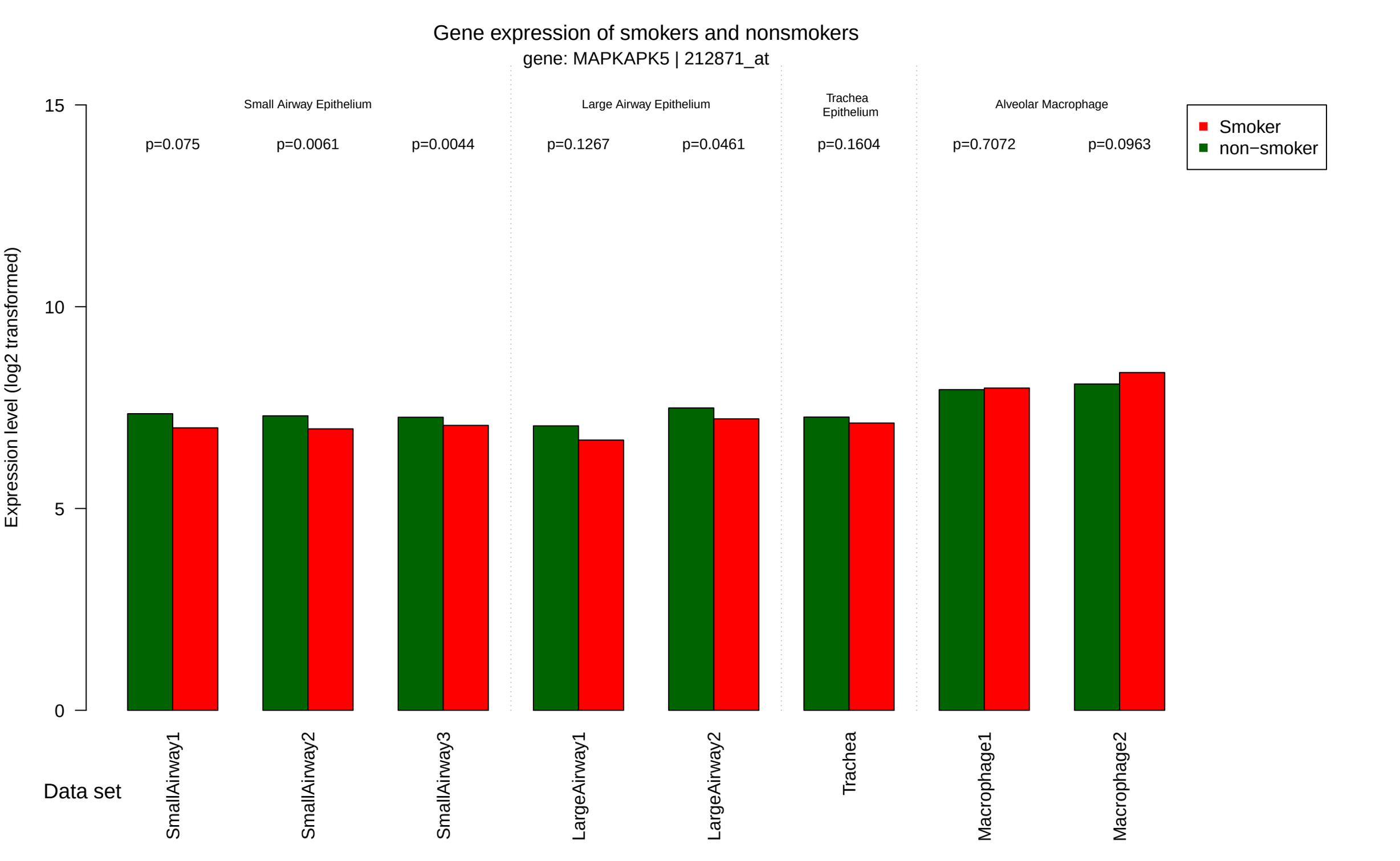
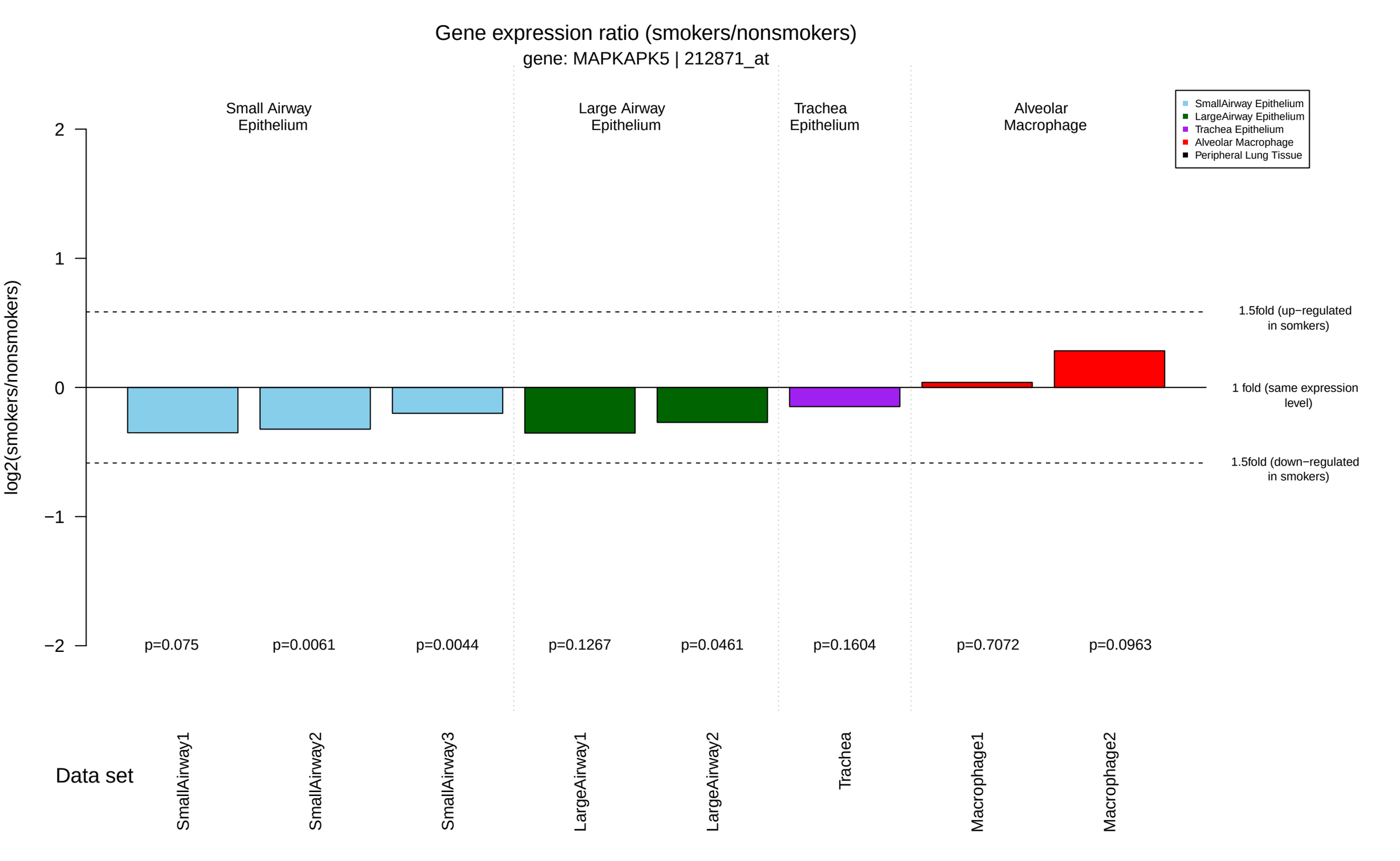
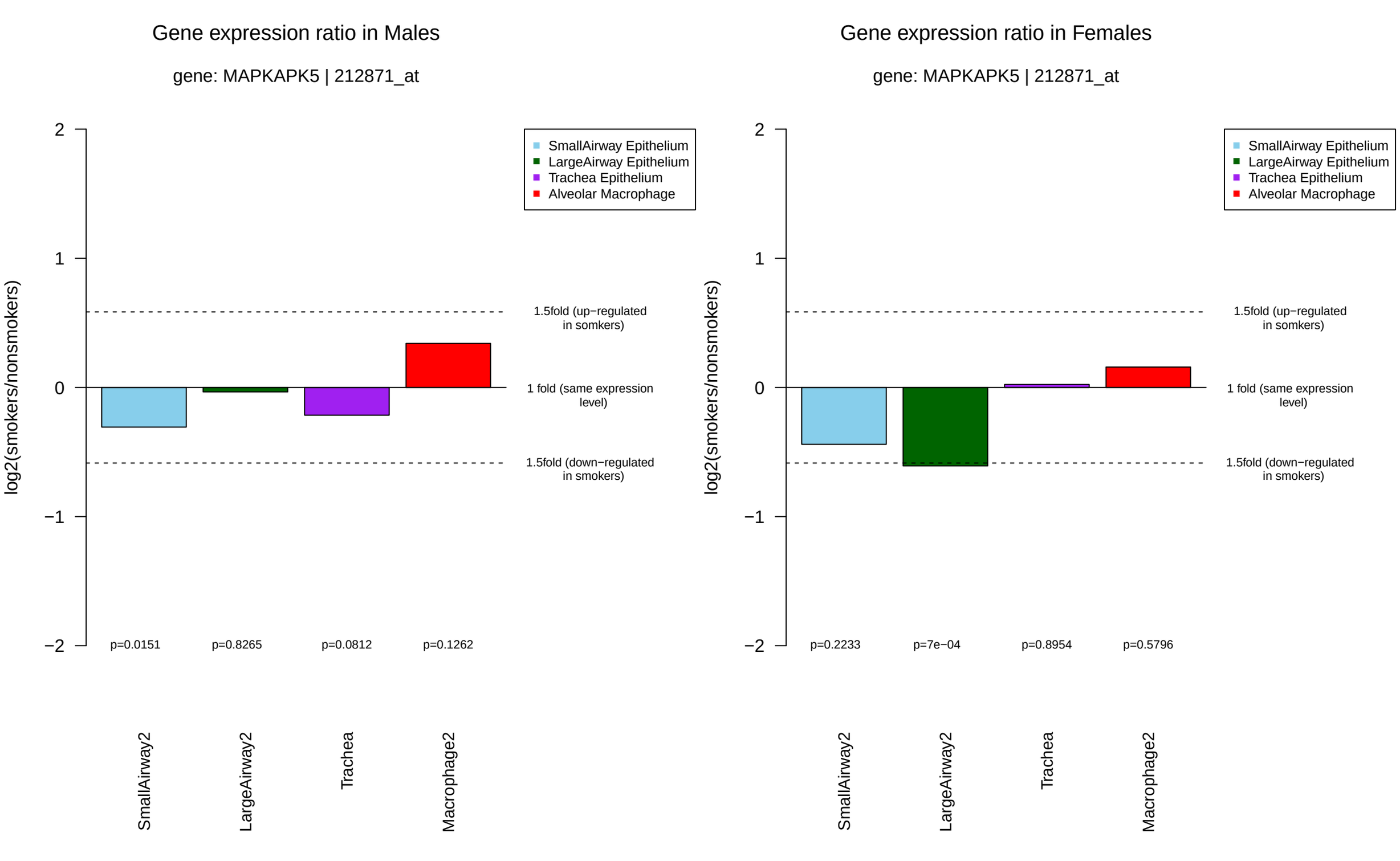
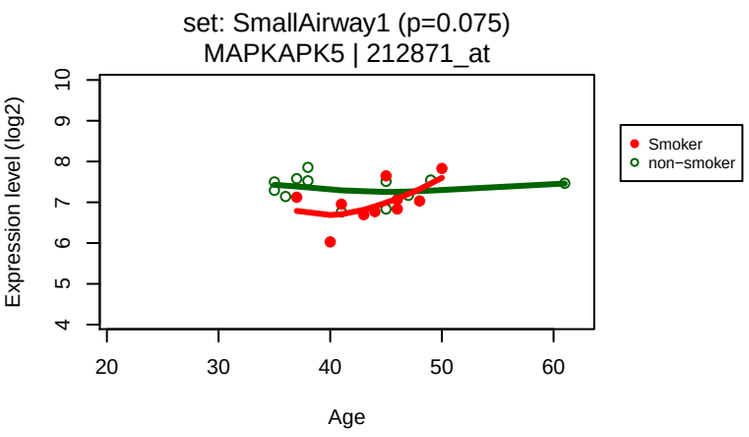
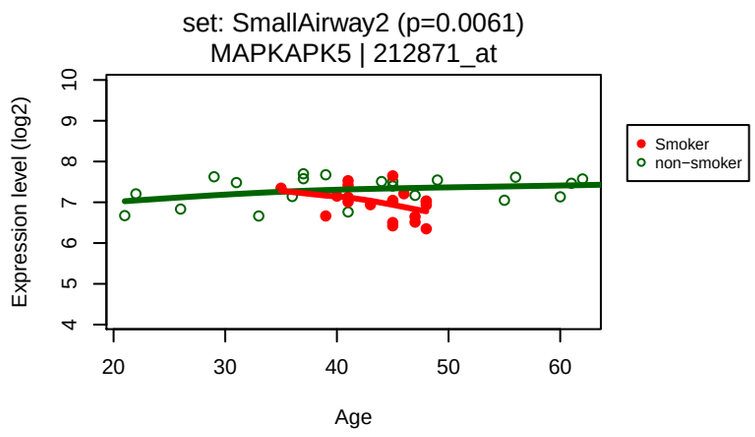
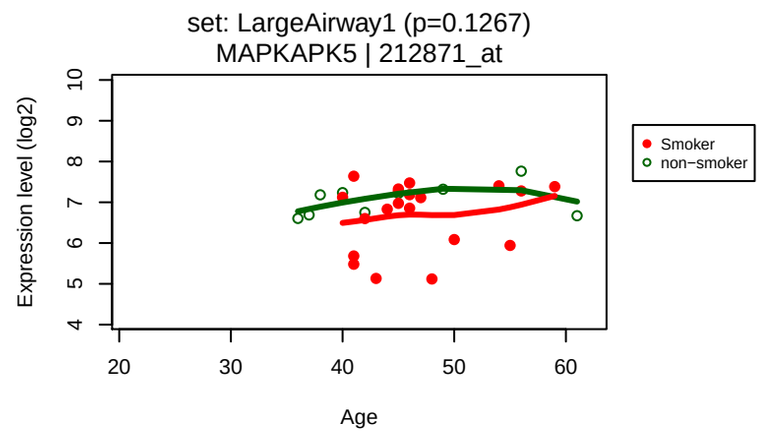

Gene expression of smokers and nonsmokers
gene: MAPKAPK5 | 212871_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.075
p=0.0061
p=0.0044
p=0.1267
p=0.0461
p=0.1604
p=0.7072
p=0.0963
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

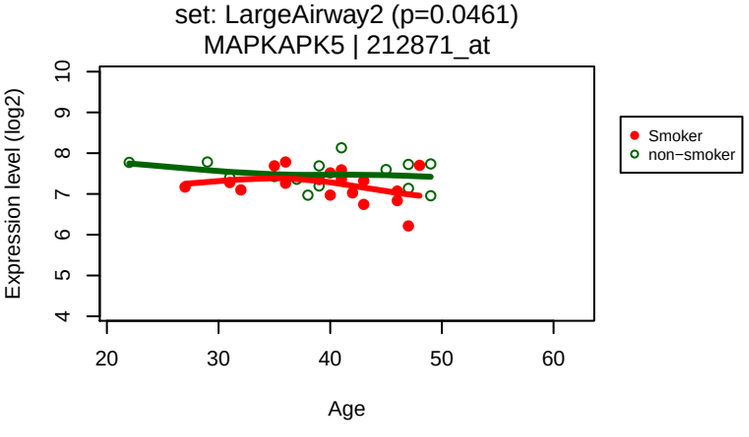

Gene expression ratio (smokers/nonsmokers)
gene: MAPKAPK5 | 212871_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.075
p=0.0061
p=0.0044
p=0.1267
p=0.0461
p=0.1604
p=0.7072
p=0.0963
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

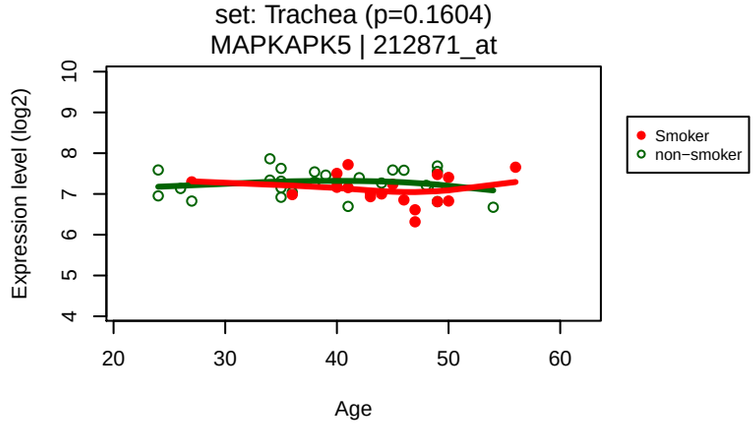

Gene expression ratio in Males
gene: MAPKAPK5 | 212871_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0151
p=0.8265
p=0.0812
p=0.1262
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPKAPK5 | 212871_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2233
p=7e−04
p=0.8954
p=0.5796
SmallAirway2
LargeAirway2
Trachea
Macrophage2

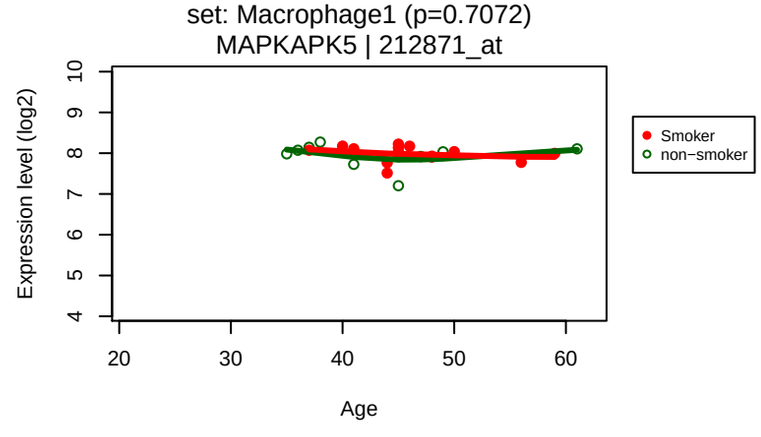

set: SmallAirway1 (p=0.075)
MAPKAPK5 | 212871_at
Expression level (log2)
Age
Smoker
non−smoker

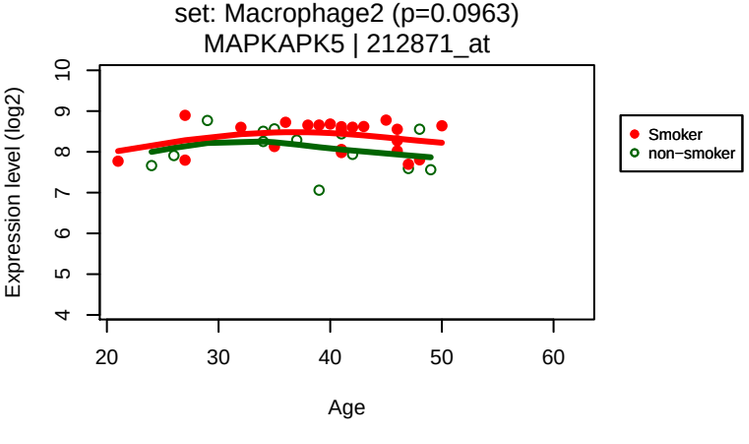

set: SmallAirway2 (p=0.0061)
MAPKAPK5 | 212871_at
Expression level (log2)
Age
Smoker
non−smoker

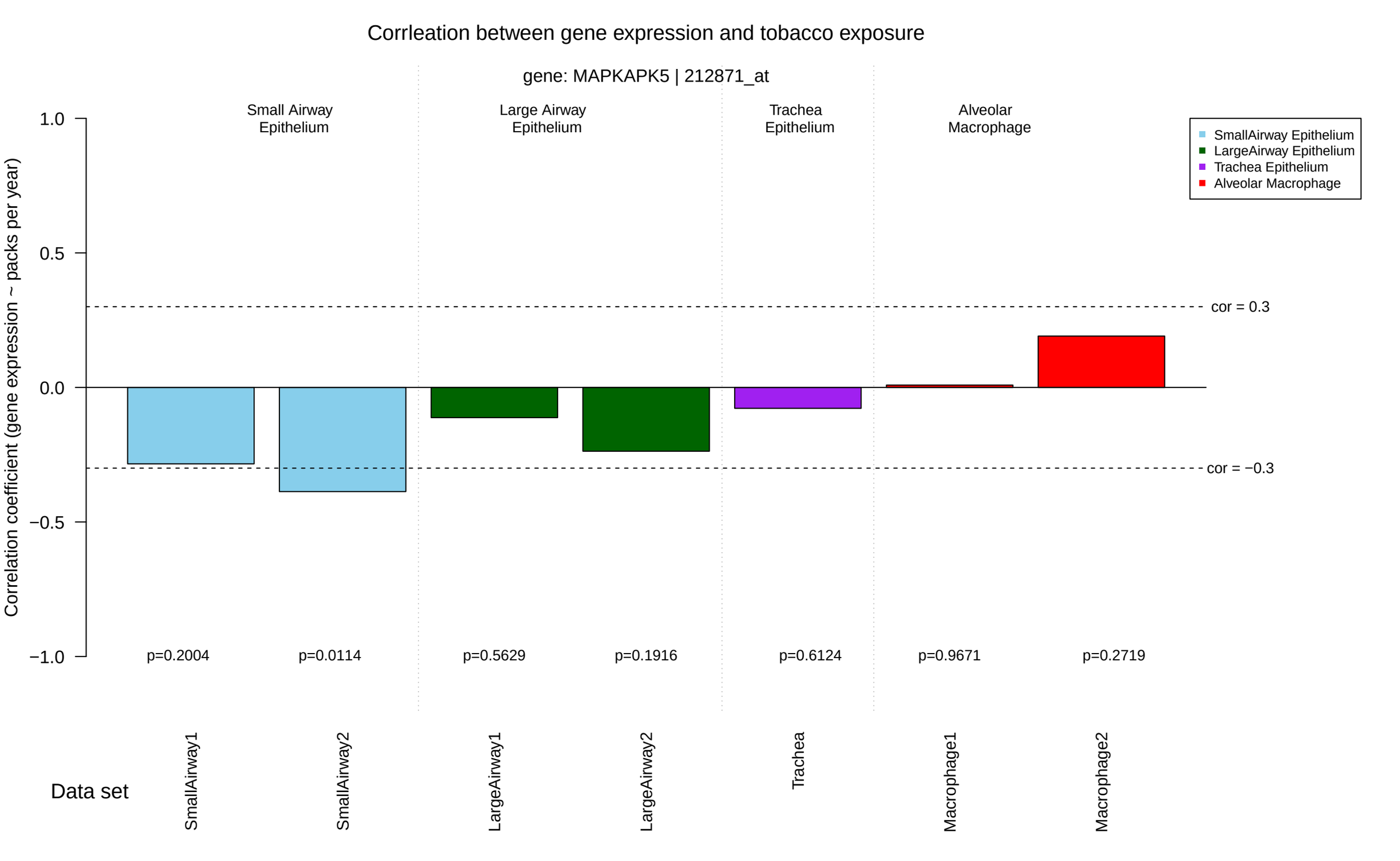

set: LargeAirway1 (p=0.1267)
MAPKAPK5 | 212871_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0461)
MAPKAPK5 | 212871_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1604)
MAPKAPK5 | 212871_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7072)
MAPKAPK5 | 212871_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0963)
MAPKAPK5 | 212871_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPKAPK5 | 212871_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2004
p=0.0114
p=0.5629
p=0.1916
p=0.6124
p=0.9671
p=0.2719
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2